

Network Routing Technologies

- To provide basics of Network Routing Technologies from a Network Administrator’s

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5261 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5261.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Information Technology
Course code	5261
Course title	Network Routing Technologies
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

28 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide basics of Network Routing Technologies from a Network Administrator's point of view.
- To provide the basics of a Network Router and its Administration.

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying and Inter-VLAN Routing for a simple network. Identify how to configure EIGRP and OSPF routing	See official syllabus
CO2	14 Applying protocol for a simple network Identify how to configure Multi-Area OSPF and IPv6	See official syllabus
CO3	15 Applying Routing for a simple network Identify the concepts of Wide Area Network and the	See official syllabus
CO4	14 Applying Evolution of Intelligent Network.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Apply basic concepts of IP Routing, Routing Protocols and Inter-VLAN	6	As specified
Module 2	network	5	As specified
Module 3	network	8	As specified
Module 4	Network.	9	As specified

MODULE 1

Apply basic concepts of IP Routing, Routing Protocols and Inter-VLAN

This module is presented from the official syllabus scope for Network Routing Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply basic concepts of IP Routing, Routing Protocols and Inter-VLAN
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and

Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and is an official topic in Module 1 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the ip routing process 2 understanding identify how to configure static routing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓൗട്ലൈൻ ഐ.ആർ.റൂട്ടിംഗ് പ്രോസസ്സ് 2 Understanding Identify how to configure Static Routing and ഐ.ആർ.റൂട്ടിംഗ് പ്രോസസ്സ് definition/meaning ഐ.ആർ.റൂട്ടിംഗ് പ്രോസസ്സ്. ഐ.ആർ.റൂട്ടിംഗ് പ്രോസസ്സ് steps, application ഐ.ആർ.റൂട്ടിംഗ് പ്രോസസ്സ്. Technical terms English-ഐ.ആർ.റൂട്ടിംഗ് പ്രോസസ്സ്.

3 Applying Default Routing

3 Applying Default Routing is an official topic in Module 1 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Default Routing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying default routing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Default Routing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Applying Default Routing
 definition/meaning . . . , steps, application Technical terms English-
 .

Summarize Dynamic Routing Protocols

Summarize Dynamic Routing Protocols is an official topic in Module 1 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Dynamic Routing Protocols using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize dynamic routing protocols should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Dynamic Routing Protocols means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Summarize Dynamic Routing Protocols
definition/meaning .
steps, application . Technical terms English-
.

Identify how to configure RIP

Identify how to configure RIP is an official topic in Module 1 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure RIP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure rip should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure RIP means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Identify how to configure RIP
definition/meaning
application Technical terms English-
.

Outline how routing between VLANs happens

Outline how routing between VLANs happens is an official topic in Module 1 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline how routing between VLANs happens using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline how routing between vlans happens should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline how routing between VLANs happens means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Outline how routing between VLANs happens
 ഹൈസ്കൂൾ തലം definition/meaning ഹൈസ്കൂൾ തലം. ഹൈസ്കൂൾ തലം ഹൈസ്കൂൾ,
 steps, application ഹൈസ്കൂൾ തലം ഹൈസ്കൂൾ. Technical terms English- ഹൈസ്കൂൾ
 ഹൈസ്കൂൾ തലം.

Identify how to Configure Inter-VLAN Routing 3 Applying Routing - IP Routing Process - Configuring IP Routing - Static Routing - Default Routing - Dynamic Routing - Administrative Distance - Routing Protocols - Routing Information Protocol (RIP) - Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing Identify how to configure EIGRP and OSPF routing protocol for a simple

Identify how to Configure Inter-VLAN Routing 3 Applying Routing - IP Routing Process - Configuring IP Routing - Static Routing - Default Routing - Dynamic Routing - Administrative Distance - Routing Protocols - Routing Information Protocol (RIP) - Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing Identify how to configure EIGRP and OSPF routing protocol for a simple is an official topic in Module 1 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to Configure Inter-VLAN Routing 3 Applying Routing - IP Routing Process - Configuring IP Routing - Static Routing - Default Routing - Dynamic Routing - Administrative Distance - Routing Protocols - Routing Information Protocol (RIP) - Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing Identify how to configure EIGRP and OSPF routing protocol for a simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure inter-vlan routing 3 applying routing - ip routing process - configuring ip routing - static routing - default routing - dynamic routing - administrative distance - routing protocols - routing information protocol (rip) - configuring rip routing routing between vlans - configuring inter-vlan routing identify how to configure eigrp and ospf routing protocol for a simple should be discussed with

attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to Configure Inter-VLAN Routing 3 Applying Routing – IP Routing Process – Configuring IP Routing – Static Routing – Default Routing – Dynamic Routing – Administrative Distance – Routing Protocols – Routing Information Protocol (RIP) – Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing Identify how to configure EIGRP and OSPF routing protocol for a simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Identify how to Configure Inter-VLAN Routing 3 Applying Routing – IP Routing Process – Configuring IP Routing – Static Routing – Default Routing – Dynamic Routing – Administrative Distance – Routing Protocols – Routing Information Protocol (RIP) – Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing Identify how to configure EIGRP and OSPF routing protocol for a simple definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Apply basic concepts of IP Routing, Routing Protocols and Inter-VLAN Routing for a simple network.

M1.01	Outline the IP Routing Process	2
Understanding	Identify how to configure Static Routing and	
M1.02		3
Applying	Default Routing	
M1.03	Summarize Dynamic Routing Protocols	2
Understanding	Identify how to configure RIP	3
Applying	Outline how routing between VLANs happens	2
Understanding	Identify how to Configure Inter-VLAN Routing	3
Applying	Routing – IP Routing Process – Configuring IP Routing – Static Routing – Default Routing – Dynamic Routing – Administrative Distance – Routing Protocols – Routing Information Protocol (RIP) – Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing Identify how to configure EIGRP and OSPF routing protocol for a simple	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

network

This module is presented from the official syllabus scope for Network Routing Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: network
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Summarize EIGRP Features and Operations

Summarize EIGRP Features and Operations is an official topic in Module 2 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize EIGRP Features and Operations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize eigrp features and operations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize EIGRP Features and Operations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Summarize EIGRP Features and Operations
definition/meaning . . ,
steps, application . Technical terms English-
.

Identify how to configure EIGRP

Identify how to configure EIGRP is an official topic in Module 2 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure EIGRP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure eigrp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure EIGRP means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Identify how to configure EIGRP
definition/meaning . . . , steps,
application Technical terms English-
.

Outline OSPF Routing Protocol

Outline OSPF Routing Protocol is an official topic in Module 2 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline OSPF Routing Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline ospf routing protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline OSPF Routing Protocol means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓൗട്ലൈൻ OSPF Routing Protocol ഓപ്പറേഷനുകൾ
ഓപ്പറേഷൻ definition/meaning ഓപ്പറേഷനുകൾ. ഓപ്പറേഷൻ ഓപ്പറേഷൻ ഓപ്പറേഷൻ, steps,
application ഓപ്പറേഷൻ ഓപ്പറേഷനുകൾ ഓപ്പറേഷൻ. Technical terms English-ഓപ്പറേഷൻ
ഓപ്പറേഷനുകൾ.

Identify how to configure OSPF

Identify how to configure OSPF is an official topic in Module 2 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure OSPF using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure ospf should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure OSPF means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓ ഓ Identify how to configure OSPF ഓഓഓഓഓഓഓഓഓഓ
ഓഓഓഓ definition/meaning ഓഓഓഓഓഓഓഓഓഓ. ഓഓഓഓ ഓഓഓഓ ഓഓഓഓ, steps,
application ഓഓഓഓ ഓഓഓഓഓഓഓ ഓഓഓഓ. Technical terms English-ഓ ഓഓഓഓ
ഓഓഓഓഓഓഓഓ.

Identify how to configure Loopback Interfaces 3 Applying Enhanced IGRP - EIGRP Features and Operations - Neighbour Discovery - Reliable Transport Protocol - Diffusing Update Algorithm - Route Discovery and Maintenance - Configuring EIGRP - VLSM support and summarization - Controlling EIGRP Traffic - Split Horizon Open Shortest Path First - OSPF Basics - Terminology - Operation - LSA Flooding - SPF Tree calculation - OSPF Metrics - Configuring OSPF - Enabling - Configuring Area - Loopback Interfaces - Configuring Loopback Interfaces Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple

Identify how to configure Loopback Interfaces 3 Applying Enhanced IGRP - EIGRP Features and Operations - Neighbour Discovery - Reliable Transport Protocol - Diffusing Update Algorithm - Route Discovery and Maintenance - Configuring EIGRP - VLSM support and summarization - Controlling EIGRP Traffic - Split Horizon Open Shortest Path First - OSPF Basics - Terminology - Operation - LSA Flooding - SPF Tree calculation - OSPF Metrics - Configuring OSPF - Enabling - Configuring Area - Loopback Interfaces - Configuring Loopback Interfaces Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple is an official topic in Module 2 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure Loopback Interfaces 3 Applying Enhanced IGRP - EIGRP Features and Operations - Neighbour Discovery - Reliable Transport Protocol - Diffusing Update Algorithm - Route Discovery and Maintenance - Configuring EIGRP - VLSM support and summarization - Controlling EIGRP Traffic - Split Horizon Open Shortest Path First - OSPF Basics - Terminology - Operation - LSA Flooding - SPF Tree calculation - OSPF Metrics - Configuring OSPF - Enabling - Configuring Area - Loopback Interfaces - Configuring Loopback Interfaces Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure loopback interfaces 3 applying enhanced igrp – eigrp features and operations – neighbour discovery – reliable transport protocol – diffusing update algorithm – route discovery and maintenance – configuring eigrp – vlsn support and summarization – controlling eigrp traffic – split horizon open shortest path first – ospf basics – terminology – operation – lsa flooding – spf tree calculation – ospf metrics – configuring ospf – enabling – configuring area – loopback interfaces – configuring loopback interfaces identify how to configure multi-area ospf and ipv6 routing for a simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure Loopback Interfaces 3 Applying Enhanced IGRP – EIGRP Features and Operations – Neighbour Discovery – Reliable Transport Protocol – Diffusing Update Algorithm – Route Discovery and Maintenance – Configuring EIGRP – VLSM support and summarization – Controlling EIGRP Traffic – Split Horizon Open Shortest Path First – OSPF Basics – Terminology – Operation – LSA Flooding – SPF Tree calculation – OSPF Metrics – Configuring OSPF – Enabling – Configuring Area – Loopback Interfaces – Configuring Loopback Interfaces Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Identify how to configure Loopback Interfaces 3 Applying Enhanced IGRP – EIGRP Features and Operations – Neighbour Discovery – Reliable Transport Protocol – Diffusing Update Algorithm – Route Discovery and Maintenance – Configuring EIGRP – VLSM support and summarization – Controlling EIGRP Traffic – Split Horizon Open Shortest Path First – OSPF Basics – Terminology – Operation – LSA Flooding – SPF Tree calculation – OSPF Metrics – Configuring OSPF – Enabling – Configuring Area – Loopback Interfaces – Configuring Loopback Interfaces Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple definition/meaning.

steps, application ഏകദേശം ഏകദേശമായി പറയാം. Technical terms English-ൽ പറയാം
ഏകദേശമായി.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

network		
M2.01	Summarize EIGRP Features and Operations	3
Understanding		
M2.02	Identify how to configure EIGRP	3
Applying		
M2.03	Outline OSPF Routing Protocol	2
Understanding		
M2.04	Identify how to configure OSPF	3
Applying		
M2.05	Identify how to configure Loopback Interfaces	3
Applying		
	Series Test – I	1
Contents:		
Enhanced IGRP – EIGRP Features and Operations – Neighbour Discovery – Reliable		
Transport		
Protocol – Diffusing Update Algorithm – Route Discovery and Maintenance –		
Configuring EIGRP		
– VLSM support and summarization – Controlling EIGRP Traffic – Split Horizon		
Open Shortest Path First – OSPF Basics – Terminology – Operation – LSA Flooding –		
SPF Tree		
calculation – OSPF Metrics – Configuring OSPF – Enabling – Configuring Area –		
Loopback		
Interfaces – Configuring Loopback Interfaces		
	Identify how to configure Multi-Area OSPF and IPv6 Routing for a	
simple		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

network

This module is presented from the official syllabus scope for Network Routing Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: network
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area

Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline categories of multi-area components 1 understanding identify how to configure basic multi-area should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area
 definition/meaning .
 , steps, application
 . Technical terms English-

3 Applying OSPF

3 Applying OSPF is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying OSPF using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying ospf should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying OSPF means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Applying OSPF സ്പെസിഫിക്കേഷൻ ഓപ്പറേഷൻ definition/meaning സ്പെസിഫൈ ചെയ്തത്. ഓപ്പറേഷൻ ഓപ്പറേഷൻ ഓപ്പറേഷൻ, steps, application ഓപ്പറേഷൻ ഓപ്പറേഷൻ ഓപ്പറേഷൻ. Technical terms English- സ്പെസിഫൈ ചെയ്തത്.

Outline needs and features of IPv6 1 Understanding Summarize Hexadecimal number system, IPv6

Outline needs and features of IPv6 1 Understanding Summarize Hexadecimal number system, IPv6 is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline needs and features of IPv6 1 Understanding Summarize Hexadecimal number system, IPv6 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline needs and features of ipv6 1 understanding summarize hexadecimal number system, ipv6 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline needs and features of IPv6 1 Understanding Summarize Hexadecimal number system, IPv6 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓൗട്ട്ലൈൻ നീക്കം ചെയ്ത ഐപിv6 1

Understanding Summarize Hexadecimal number system, IPv6 ഐപിv6 നമ്പർ സിസ്റ്റം ഐപിv6 definition/meaning ഐപിv6 നമ്പർ സിസ്റ്റം. ഐപിv6 നമ്പർ സിസ്റ്റം, steps, application ഐപിv6 നമ്പർ സിസ്റ്റം. Technical terms English-ഐപിv6 നമ്പർ സിസ്റ്റം.

1 Understanding notation and prefixes

1 Understanding notation and prefixes is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding notation and prefixes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding notation and prefixes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding notation and prefixes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-1 Understanding notation and prefixes
പ്രതീകരണവും പ്രതീകരണത്തിന്റെ definition/meaning ഉൾപ്പെടെയും. പ്രതീകരണത്തിന്റെ പ്രയോഗം, steps, application ഉൾപ്പെടെയും. Technical terms English-ൽ പ്രതീകരണത്തിന്റെ പ്രയോഗം.

Illustrate IPv6 Subnets and addressing types

Illustrate IPv6 Subnets and addressing types is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate IPv6 Subnets and addressing types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate ipv6 subnets and addressing types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate IPv6 Subnets and addressing types means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- IPv6 Subnets and addressing types
IPv6 Subnets and addressing types definition/meaning. IPv6 Subnets and addressing types, steps, application. Technical terms English- Malayalam
translation.

Outline the working of IPv6 in an internetwork

Outline the working of IPv6 in an internetwork is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the working of IPv6 in an internetwork using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the working of ipv6 in an internetwork should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the working of IPv6 in an internetwork means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- IPv6 Outline the working of IPv6 in an internetwork
IPv6 definition/meaning IPv6. IPv6 steps, application IPv6. Technical terms English-
IPv6.

Summarize IPv6 routing protocols 2 Understanding Identify how to configure IPv6 Routing for a

Summarize IPv6 routing protocols 2 Understanding Identify how to configure IPv6 Routing for a is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize IPv6 routing protocols 2 Understanding Identify how to configure IPv6 Routing for a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize ipv6 routing protocols 2 understanding identify how to configure ipv6 routing for a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize IPv6 routing protocols 2 Understanding Identify how to configure IPv6 Routing for a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Summarize IPv6 routing protocols 2

Understanding Identify how to configure IPv6 Routing for a **network** **definition/meaning** **network**. **network** **network** **network**, **steps**, **application** **network** **network** **network**. **Technical terms** **English-
network** **network**.

3 Applying simple network Multi-Area OSPF - OSPF Scalability - Categories of Multi-Area Components - Adjacency Requirements - OSPF Router Roles - Link-State Advertisements - Hello Protocol - Neighbour States - Basic Multi-Area Configuration IPv6 - Need - Features - Hexadecimal Number System - IPv6 Addressing Notation - IPv6 Prefixes - IPv6 Subnets - IPv6 Addressing Types - Unicast - Multicast - Anycast - Working of IPv6 in an internetwork - Manual Address assignment - Stateless Autoconfiguration - DHCPv6 - IPv6 Routing Protocols - Static Routing with IPv6 - Configuring IPv6 Routing Identify the concepts of Wide Area Network and the Evolution of Intelligent

3 Applying simple network Multi-Area OSPF - OSPF Scalability - Categories of Multi-Area Components - Adjacency Requirements - OSPF Router Roles - Link-State Advertisements - Hello Protocol - Neighbour States - Basic Multi-Area Configuration IPv6 - Need - Features - Hexadecimal Number System - IPv6 Addressing Notation - IPv6 Prefixes - IPv6 Subnets - IPv6 Addressing Types - Unicast - Multicast - Anycast - Working of IPv6 in an internetwork - Manual Address assignment - Stateless Autoconfiguration - DHCPv6 - IPv6 Routing Protocols - Static Routing with IPv6 - Configuring IPv6 Routing Identify the concepts of Wide Area Network and the Evolution of Intelligent is an official topic in Module 3 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying simple network Multi-Area OSPF - OSPF Scalability - Categories of Multi-Area Components - Adjacency Requirements - OSPF Router Roles - Link-State Advertisements - Hello Protocol - Neighbour States - Basic Multi-Area Configuration IPv6 - Need - Features - Hexadecimal Number System - IPv6 Addressing Notation - IPv6 Prefixes - IPv6 Subnets - IPv6 Addressing Types - Unicast - Multicast - Anycast - Working of IPv6 in an internetwork - Manual Address assignment - Stateless Autoconfiguration - DHCPv6 - IPv6 Routing Protocols - Static Routing with IPv6 - Configuring IPv6 Routing Identify the concepts of Wide Area Network and the Evolution of Intelligent using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying simple network multi-area ospf – ospf scalability – categories of multi-area components – adjacency requirements – ospf router roles – link-state advertisements – hello protocol – neighbour states – basic multi-area configuration ipv6 – need – features – hexadecimal number system – ipv6 addressing notation – ipv6 prefixes – ipv6 subnets – ipv6 addressing types – unicast – multicast – anycast – working of ipv6 in an internetwork – manual address assignment – stateless autoconfiguration – dhcpv6 – ipv6 routing protocols – static routing with ipv6 – configuring ipv6 routing identify the concepts of wide area network and the evolution of intelligent should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying simple network Multi-Area OSPF – OSPF Scalability – Categories of Multi-Area Components – Adjacency Requirements – OSPF Router Roles – Link-State Advertisements – Hello Protocol – Neighbour States – Basic Multi-Area Configuration IPv6 – Need – Features – Hexadecimal Number System – IPv6 Addressing Notation – IPv6 Prefixes – IPv6 Subnets – IPv6 Addressing Types – Unicast – Multicast – Anycast – Working of IPv6 in an internetwork – Manual Address assignment – Stateless Autoconfiguration – DHCPv6 – IPv6 Routing Protocols – Static Routing with IPv6 – Configuring IPv6 Routing Identify the concepts of Wide Area Network and the Evolution of Intelligent means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Applying simple network Multi-Area OSPF – OSPF Scalability – Categories of Multi-Area Components – Adjacency Requirements – OSPF Router Roles – Link-State Advertisements – Hello Protocol – Neighbour States – Basic Multi-Area Configuration IPv6 – Need – Features – Hexadecimal Number System – IPv6 Addressing Notation – IPv6 Prefixes – IPv6 Subnets – IPv6 Addressing Types – Unicast – Multicast – Anycast – Working of IPv6 in an

internetwork – Manual Address assignment – Stateless Autoconfiguration – DHCPv6
 – IPv6 Routing Protocols – Static Routing with IPv6 – Configuring IPv6 Routing
 Identify the concepts of Wide Area Network and the Evolution of Intelligent
 നെറ്റ്വർക്ക് നെറ്റ്വർക്ക് definition/meaning നെറ്റ്വർക്ക് നെറ്റ്വർക്ക്, steps, application നെറ്റ്വർക്ക് നെറ്റ്വർക്ക് നെറ്റ്വർക്ക്. Technical terms English-നെറ്റ്വർക്ക് നെറ്റ്വർക്ക്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
 Educational explanations below are separate elaboration.

network		
M3.01	Outline categories of Multi-Area components	1
Understanding	Identify how to configure basic multi-area	
M3.02		3
Applying	OSPF	
M3.03	Outline needs and features of IPv6	1
Understanding	Summarize Hexadecimal number system, IPv6	
M3.04		1
Understanding	notation and prefixes	
M3.05	Illustrate IPv6 Subnets and addressing types	2
Understanding		
M3.06	Outline the working of IPv6 in an internetwork	2
Understanding		
M3.07	Summarize IPv6 routing protocols	2
Understanding	Identify how to configure IPv6 Routing for a	
M3.08		3
Applying	simple network	
Contents:		
Multi-Area OSPF – OSPF Scalability – Categories of Multi-Area Components –		
Adjacency		
Requirements – OSPF Router Roles – Link-State Advertisements – Hello Protocol –		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Network.

This module is presented from the official syllabus scope for Network Routing Technologies. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Network.
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

Summarize Wide Area Networks

Summarize Wide Area Networks is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Wide Area Networks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize wide area networks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Wide Area Networks means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Summarize Wide Area Networks
 definition/meaning . . . , steps, application Technical terms English-
 .

Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2

Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2 is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline cabling the serial wide area network 1 understanding illustrate virtual private networks and gre 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഔ Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2 ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം ഉൾപ്പെടെ. ഓരോന്നിനും ഓരോന്നിനും, steps, application ഉൾപ്പെടെ ഓരോന്നിനും ഓരോന്നിനും. Technical terms English-ഔ ഓരോന്നിനും ഓരോന്നിനും.

Understanding Tunnels

Understanding Tunnels is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding Tunnels using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding tunnels should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding Tunnels means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന Understanding Tunnels എന്നതിന്റെ definition/meaning എന്താണ്. എന്തിന് എന്തിന്, steps, application എന്നിവ എന്തിന് എന്തിന്. Technical terms English-ൽ എന്തിന് എന്തിന്.

Identify how to configure single-homed EBG

Identify how to configure single-homed EBG is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify how to configure single-homed EBG using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify how to configure single-homed ebg should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify how to configure single-homed EBG means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഈ Identify how to configure single-homed EBGP
ഈ definition/meaning ഈ. ഈ ഈ, steps, application ഈ. Technical terms English-ഈ ഈ ഈ.

Outline Switch Stacking 1 Understanding Outline Cloud Computing and its effect on the 2

Outline Switch Stacking 1 Understanding Outline Cloud Computing and its effect on the 2 is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Switch Stacking 1 Understanding Outline Cloud Computing and its effect on the 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline switch stacking 1 understanding outline cloud computing and its effect on the 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Switch Stacking 1 Understanding Outline Cloud Computing and its effect on the 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഐ Outline Switch Stacking 1 Understanding Outline Cloud Computing and its effect on the 2 ഐഐഐഐഐഐഐഐ ഐഐഐ definition/meaning ഐഐഐഐഐഐഐഐ. ഐഐഐഐ ഐഐഐഐ ഐഐഐഐ, steps, application ഐഐഐഐ ഐഐഐഐഐഐ ഐഐഐഐ. Technical terms English-ഐ ഐഐഐഐ ഐഐഐഐഐഐഐഐ.

Understanding Enterprise Network Service Models Outline the Network Programmability in 2

Understanding Enterprise Network Service Models Outline the Network Programmability in 2 is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding Enterprise Network Service Models Outline the Network Programmability in 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding enterprise network service models outline the network programmability in 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding Enterprise Network Service Models Outline the Network Programmability in 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിട്ടുള്ള Understanding Enterprise Network Service Models Outline the Network Programmability in 2 ഏകദേശം 200 വാക്കുകളിൽ നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

Understanding Enterprise Network

Understanding Enterprise Network is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding Enterprise Network using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding enterprise network should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding Enterprise Network means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിട്ടുള്ള Understanding Enterprise Network

എന്നിട്ടുള്ള definition/meaning എന്നിട്ടുള്ള. എന്നിട്ടുള്ള എന്നിട്ടുള്ള, steps, application എന്നിട്ടുള്ള എന്നിട്ടുള്ള എന്നിട്ടുള്ള. Technical terms English-എന്നിട്ടുള്ള എന്നിട്ടുള്ള എന്നിട്ടുള്ള.

Summarize Application Program Interfaces

Summarize Application Program Interfaces is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Application Program Interfaces using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize application program interfaces should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Application Program Interfaces means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നത് Summarize Application Program Interfaces
എന്നതിനെക്കുറിച്ച് നിങ്ങളുടെ definition/meaning എഴുതുക. നിങ്ങളുടെ നിങ്ങളുടെ
steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക
എഴുതുക.

Outline APIC-EM and Intelligent WAN 2 Understanding 1 Wide Area Networks - WAN Topology Options - Defining WAN Terms - WAN Connection Bandwidth - WAN Connection Types - Cabling the Serial WAN - Serial Transmission - DTE and DCE - Virtual Private Networks - Benefits of VPN - Enterprise and Provider Managed VPNs - IPsec - GRE Tunnels - Single-Homed EBGp - Configuring EBGp Evolution of Intelligent Networks - Switch Stacking - Cloud Computing and its Effect on the Enterprise Network - Cloud Service Models - Network Programmability - Application Programming Interfaces - APIC-EM - Intelligent WAN Text / Reference: T/R Book Title / Author T1 Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. R1 Elements of Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 <https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/> <https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-2-tutorials/> 3 <https://www.guru99.com/introduction-ccna.html> 4 <https://study-ccna.com/> 5 <http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php>

Outline APIC-EM and Intelligent WAN 2 Understanding 1 Wide Area Networks - WAN Topology Options - Defining WAN Terms - WAN Connection Bandwidth - WAN Connection Types - Cabling the Serial WAN - Serial Transmission - DTE and DCE - Virtual Private Networks - Benefits of VPN - Enterprise and Provider Managed VPNs - IPsec - GRE Tunnels - Single-Homed EBGp - Configuring EBGp Evolution of Intelligent Networks - Switch Stacking - Cloud Computing and its Effect on the Enterprise Network - Cloud Service Models - Network Programmability - Application Programming Interfaces - APIC-EM - Intelligent WAN Text / Reference: T/R Book Title / Author T1 Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. R1 Elements of Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 <https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/> <https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-2-tutorials/> 3 <https://www.guru99.com/introduction-ccna.html> 4 <https://study-ccna.com/> 5 <http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php> is an official topic in Module 4 of Network Routing Technologies. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline APIC-EM and Intelligent WAN 2 Understanding 1 Wide Area Networks – WAN Topology Options – Defining WAN Terms – WAN Connection Bandwidth – WAN Connection Types – Cabling the Serial WAN – Serial Transmission – DTE and DCE – Virtual Private Networks – Benefits of VPN – Enterprise and Provider Managed VPNs – IPsec – GRE Tunnels – Single-Homed EBGP – Configuring EBGP Evolution of Intelligent Networks – Switch Stacking – Cloud Computing and its Effect on the Enterprise Network – Cloud Service Models – Network Programmability – Application Programming Interfaces – APIC-EM – Intelligent WAN Text / Reference: T/R Book Title / Author T1 Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. R1 Elements of Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 <https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/> 2 <https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-2-tutorials/> 3 <https://www.guru99.com/introduction-ccna.html> 4 <https://study-ccna.com/> 5 <http://www.omniseu.com/cisco-certified-network-associate-ccna/index.php> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline apic-em and intelligent wan 2 understanding 1 wide area networks – wan topology options – defining wan terms – wan connection bandwidth – wan connection types – cabling the serial wan – serial transmission – dte and dce – virtual private networks – benefits of vpn – enterprise and provider managed vpns – ipsec – gre tunnels – single-homed ebgp – configuring ebgp evolution of intelligent networks – switch stacking – cloud computing and its effect on the enterprise network – cloud service models – network programmability – application programming interfaces – apic-em – intelligent wan text / reference: t/r book title / author t1 todd lammle, ccna routing and switching complete study guide, 2nd edition, sybex (wiley) publication, 2016 narasimha karumanchi, dr. a. damodaram and dr. m. sreenivasa rao. r1 elements of computer networking: an integrated approach. careermonk publications, 2014. online resources: sl.no website link 1 <https://www.freeccnastudyguide.com/study-guides/icnd2->

200-101/ <https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-2-tutorials/> 3 <https://www.guru99.com/introduction-ccna.html> 4 <https://study-ccna.com/> 5 <http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline APIC-EM and Intelligent WAN 2 Understanding 1 Wide Area Networks – WAN Topology Options – Defining WAN Terms – WAN Connection Bandwidth – WAN Connection Types – Cabling the Serial WAN – Serial Transmission – DTE and DCE – Virtual Private Networks – Benefits of VPN – Enterprise and Provider Managed VPNs – IPsec – GRE Tunnels – Single-Homed EBG – Configuring EBG Evolution of Intelligent Networks – Switch Stacking – Cloud Computing and its Effect on the Enterprise Network – Cloud Service Models – Network Programmability – Application Programming Interfaces – APIC-EM – Intelligent WAN Text / Reference: T/R Book Title / Author T1 Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. R1 Elements of Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1 https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/ https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-2-tutorials/ https://www.guru99.com/introduction-ccna.html 4 https://study-ccna.com/ 5 http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline APIC-EM and Intelligent WAN 2 Understanding 1 Wide Area Networks – WAN Topology Options – Defining WAN Terms – WAN Connection Bandwidth – WAN Connection Types – Cabling the Serial WAN – Serial Transmission – DTE and DCE – Virtual Private Networks – Benefits of VPN – Enterprise and Provider Managed VPNs – IPsec – GRE Tunnels – Single-Homed EBG – Configuring EBG Evolution of Intelligent Networks – Switch Stacking – Cloud Computing and its Effect on the Enterprise Network – Cloud Service Models – Network Programmability – Application Programming Interfaces – APIC-EM – Intelligent WAN Text / Reference: T/R Book Title / Author T1 Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, Sybex (Wiley) Publication, 2016 Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. R1 Elements of Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Online Resources: Sl.No Website Link 1

English-□ □□□□□ □□□□□□□□□□.

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Network.		
M4.01	Summarize Wide Area Networks	1
Understanding		
M4.02	Outline Cabling the Serial Wide Area Network	1
Understanding		
	Illustrate Virtual Private Networks and GRE	2
M4.03		
Understanding		
	Tunnels	
M4.04	Identify how to configure single-homed EBGp	2
Applying		
M4.05	Outline Switch Stacking	1
Understanding		
	Outline Cloud Computing and its effect on the	2
M4.06		
Understanding		
	Enterprise Network Service Models	
	Outline the Network Programmability in	2
M4.07		
Understanding		
	Enterprise Network	
M4.08	Summarize Application Program Interfaces	1
Understanding		
M4.09	Outline APIC-EM and Intelligent WAN	2
Understanding		
		1
	Series Test – II	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying Default Routing. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying Default Routing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize Dynamic Routing Protocols. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Dynamic Routing Protocols*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify how to configure RIP. (3 marks)

Answer: Begin with the standard definition or identity of *Identify how to configure RIP*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Summarize EIGRP Features and Operations. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize EIGRP Features and Operations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify how to configure EIGRP. (3 marks)

Answer: Begin with the standard definition or identity of *Identify how to configure EIGRP*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline OSPF Routing Protocol. (3 marks)

Answer: Begin with the standard definition or identity of *Outline OSPF Routing Protocol*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Identify how to configure OSPF. (3 marks)

Answer: Begin with the standard definition or identity of *Identify how to configure OSPF*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 3

Define or explain Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area. (3 marks)

Answer: Begin with the standard definition or identity of *Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Applying OSPF. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying OSPF*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Outline needs and features of IPv6 1 Understanding Summarize Hexadecimal number system, IPv6. (3 marks)

Answer: Begin with the standard definition or identity of *Outline needs and features of IPv6 1 Understanding Summarize Hexadecimal number system, IPv6*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 1 Understanding notation and prefixes. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding notation and prefixes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Summarize Wide Area Networks. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Wide Area Networks*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Cabling the Serial Wide Area Network 1 Understanding Illustrate Virtual Private Networks and GRE 2*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Understanding Tunnels. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding Tunnels*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Identify how to configure single-homed EBG. (3 marks)

Answer: Begin with the standard definition or identity of *Identify how to configure single-homed EBG*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Outline the IP Routing Process 2 Understanding Identify how to configure Static Routing and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Summarize EIGRP Features and Operations.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Outline categories of Multi-Area components 1 Understanding Identify how to configure basic multi-area.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Summarize Wide Area Networks**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.

Term	Meaning in this lesson
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/>
- <https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-tutorials/>
- <https://www.guru99.com/introduction-ccna.html> <https://study-ccna.com/>
- <http://www.omniseu.com/cisco-certified-network-associate-ccna/index.php>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5261. Verify institutional updates against the current official curriculum.